

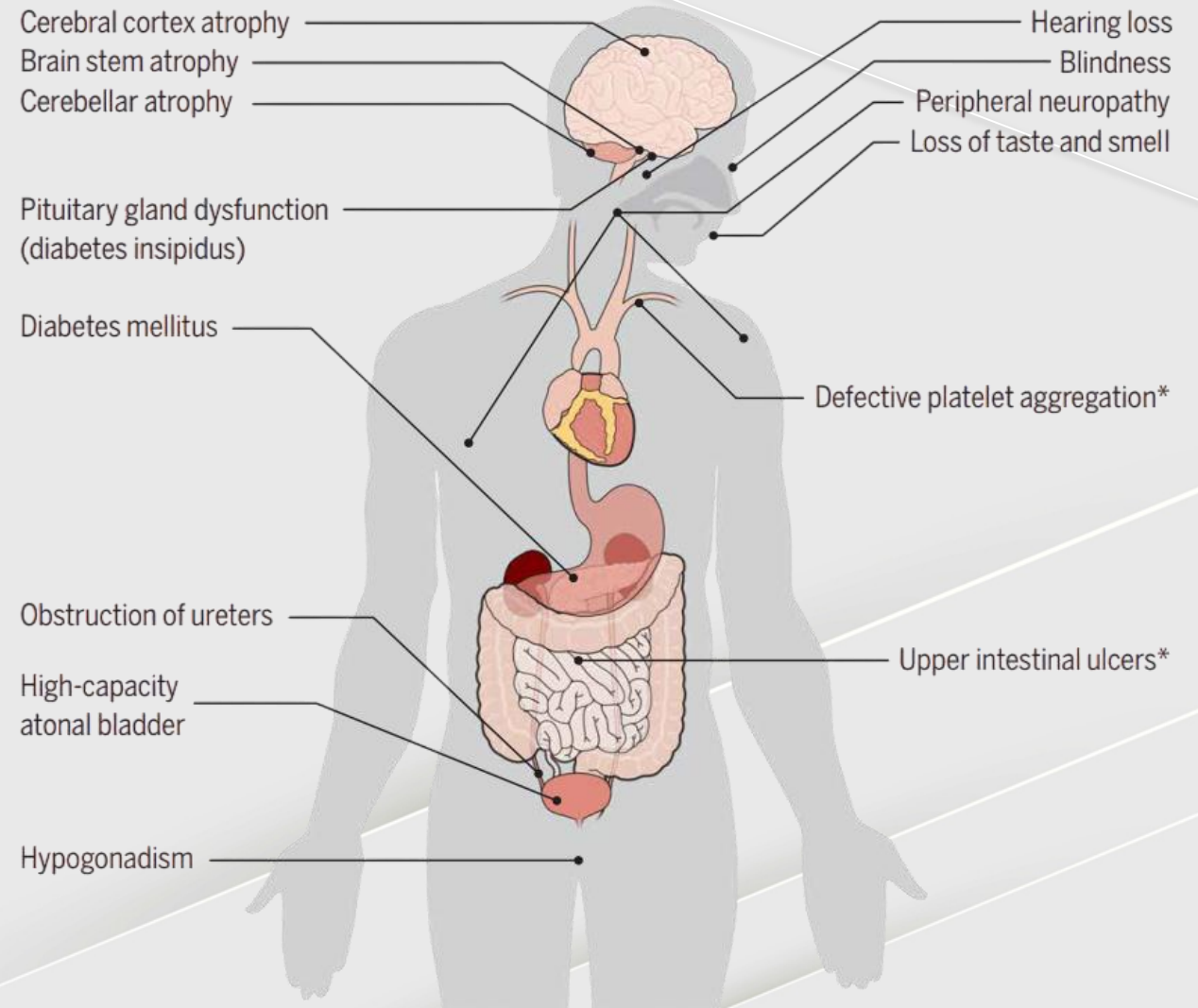


# The Role of CISO2 in RyR3-Mediated Calcium Release in Wolfram Syndrome Type 2



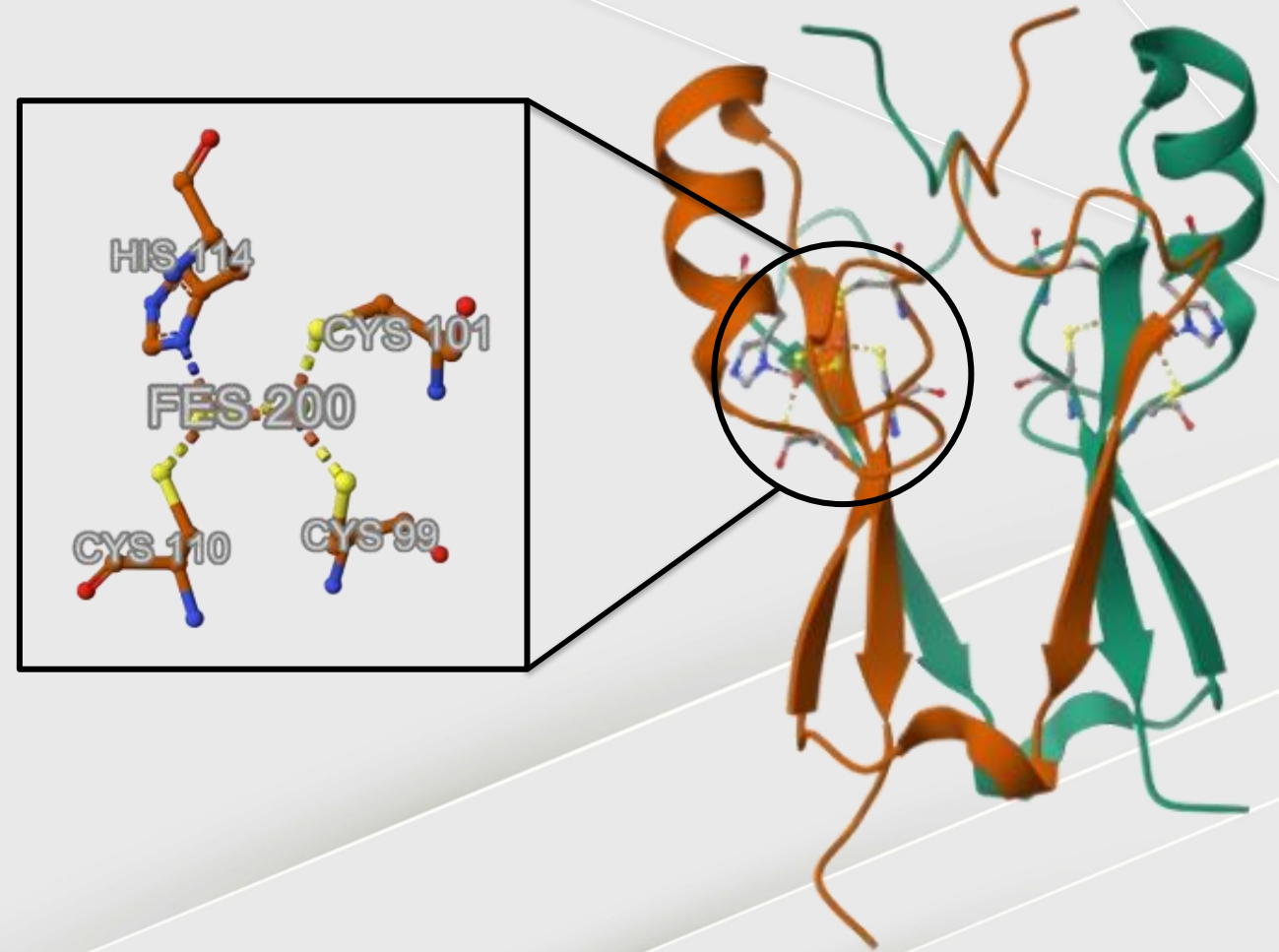
# Wolfram Syndrome

- Rare neurodegenerative disorder
- WS1 caused by WFS1 mutations
- WS2 caused by CISD2 mutations
- Life expectancy of 30 years



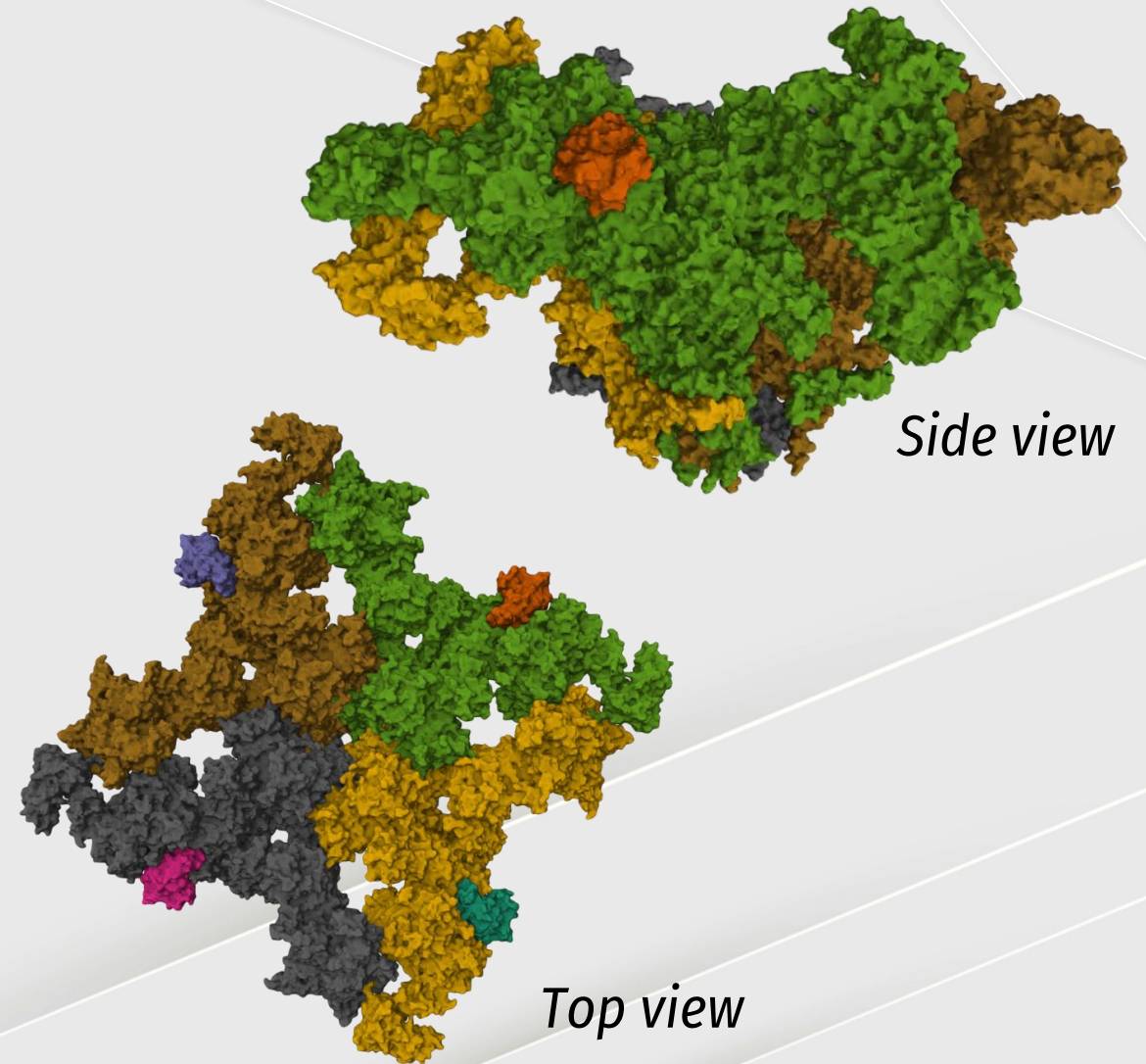
# CISD2

- Transmembrane protein enriched in MAMs
- 2Fe-2S cluster domain
- Regulates  $\text{Ca}^{2+}$  mediated processes



# Ryanodine receptors

- Largest known ion channels
- 3 isoforms (RyR1, RyR2, RyR3)
- Expressed on the ER of excitable tissues
- Agonists: ATP and caffeine

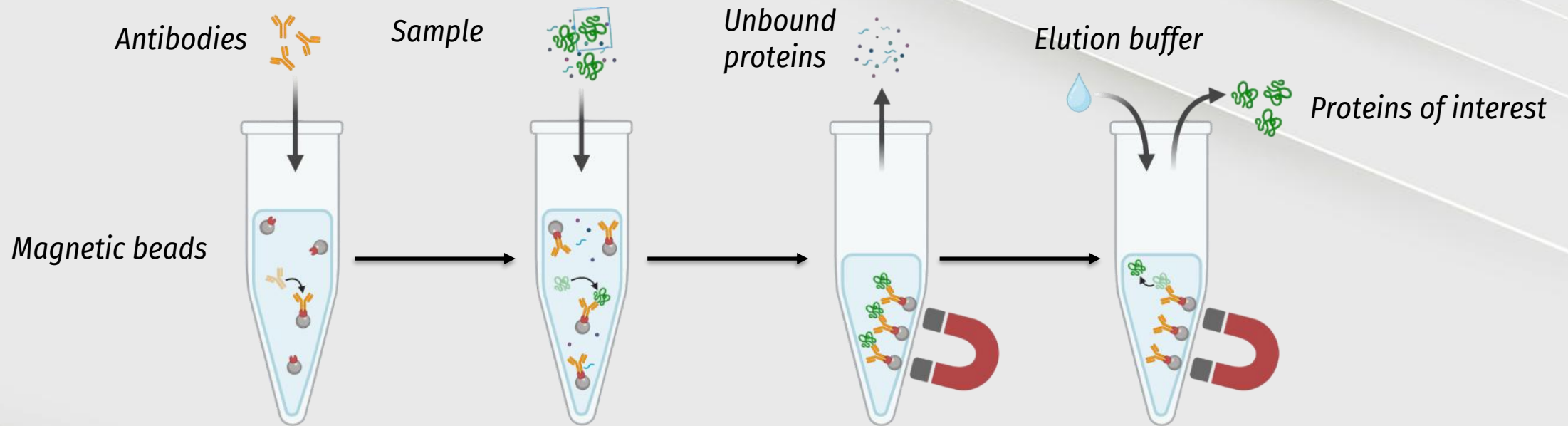


# Goals of the project

- ❑ Investigate the physical interaction between CISO2 and RyR3
  - ❑ Assess CISO2 impact on RyR3-mediated  $\text{Ca}^{2+}$  release
- ❑ Evaluate the effect of CISO2 mutants on RyR3-mediated  $\text{Ca}^{2+}$  release
- ❑ Evaluate the mitochondrial  $\text{Ca}^{2+}$  level in upon RyR3 stimulation

# **Experimental methods and techniques**

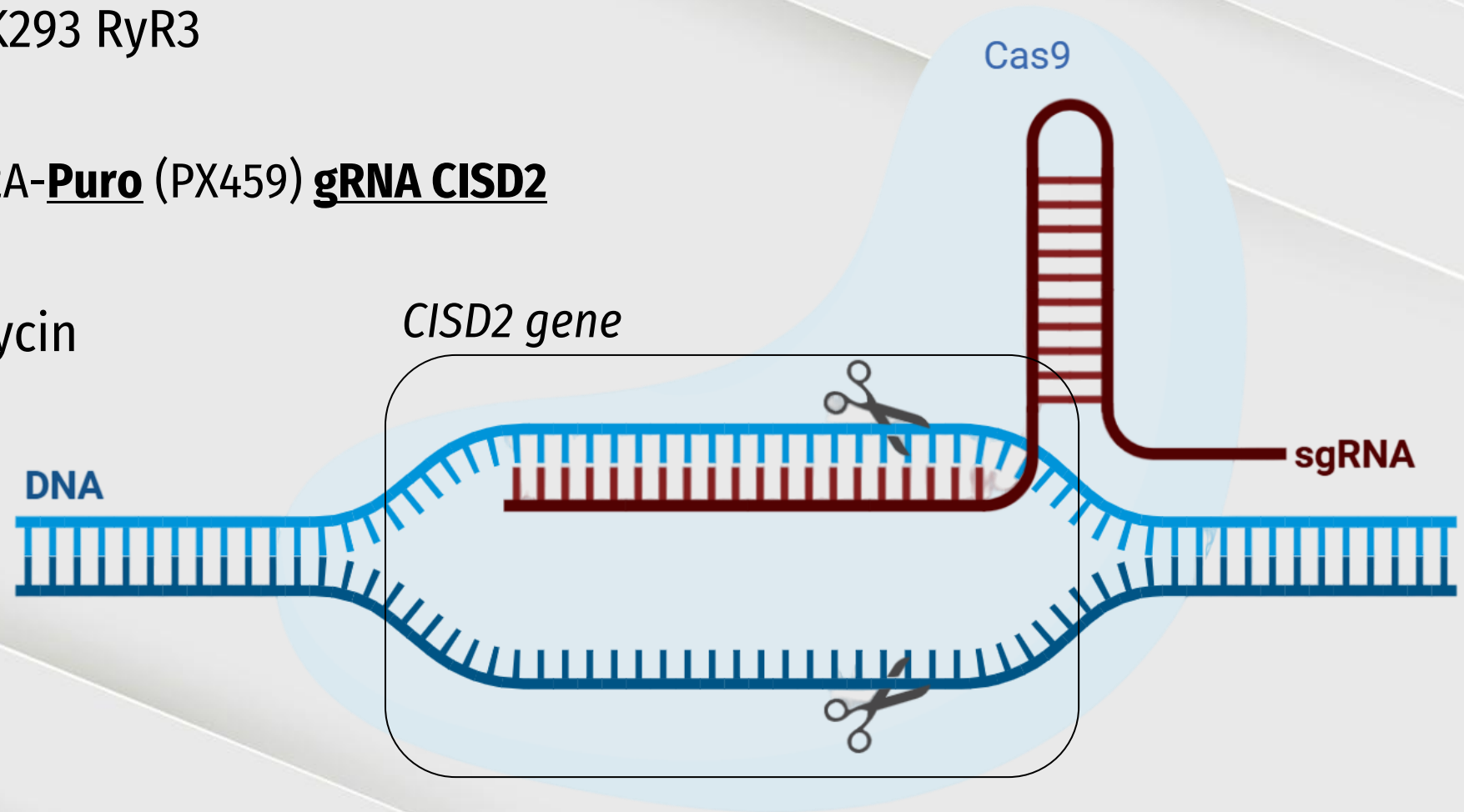
# Co-Immunoprecipitation



- Antibodies: C34 anti-RyR3
- Sample: cell lysate
- Magnetic beads: Dynabeads® Protein G
- Elution buffer: SDS based solution

# CRISPR-Cas9 genome editing

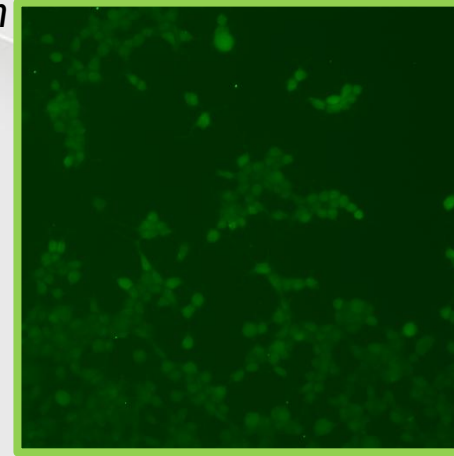
- Transfected cells: HEK293 RyR3
- Plasmid: pSpCas9(BB)-2A-Puro (PX459) gRNA C1SD2
- Cell selection: puromycin



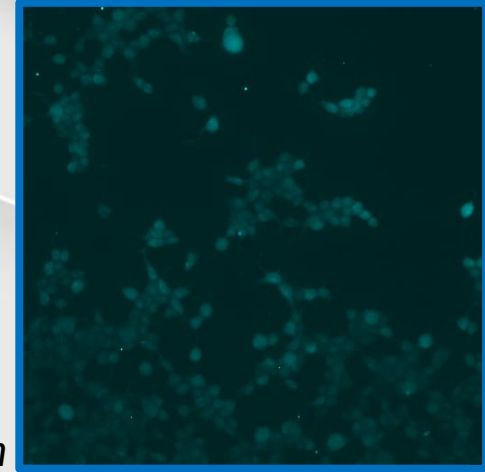
# Fluorescence microscopy

- Cytosolic and mitochondrial  $\text{Ca}^{2+}$  measurements
- FURA2-AM excited at 380 or 340 nm
- mCherry excited at 550 nm
- Green mtCEPIA excited at 470 nm

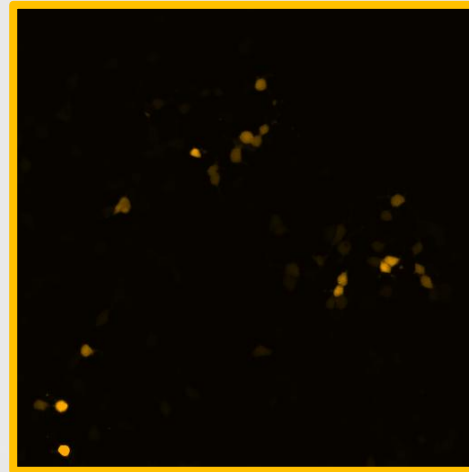
*FURA2 340 nm*



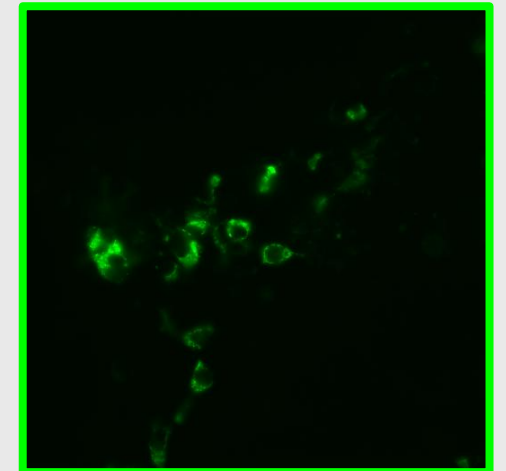
*FURA2 380 nm*



*mCherry*



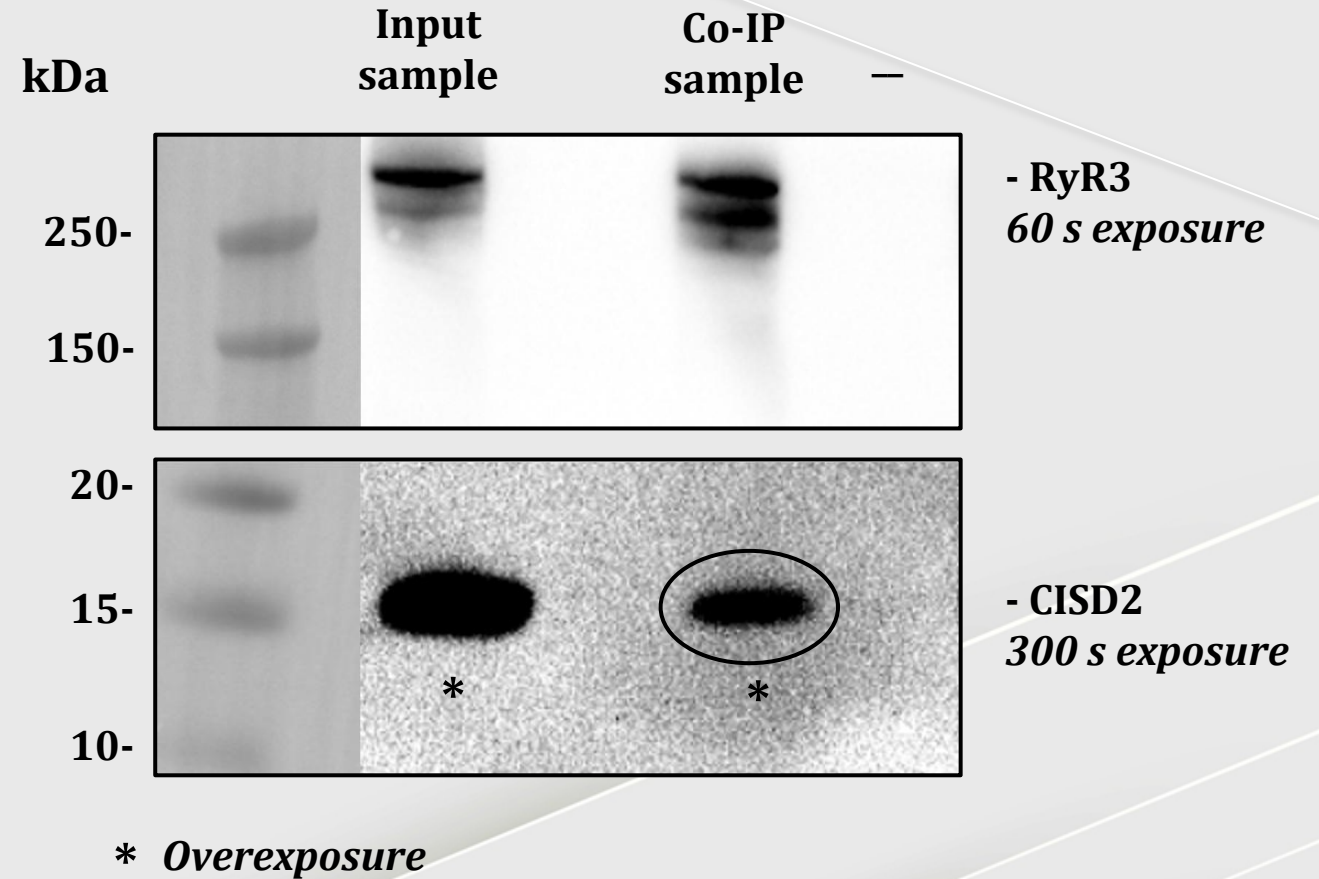
*mtCEPIA*



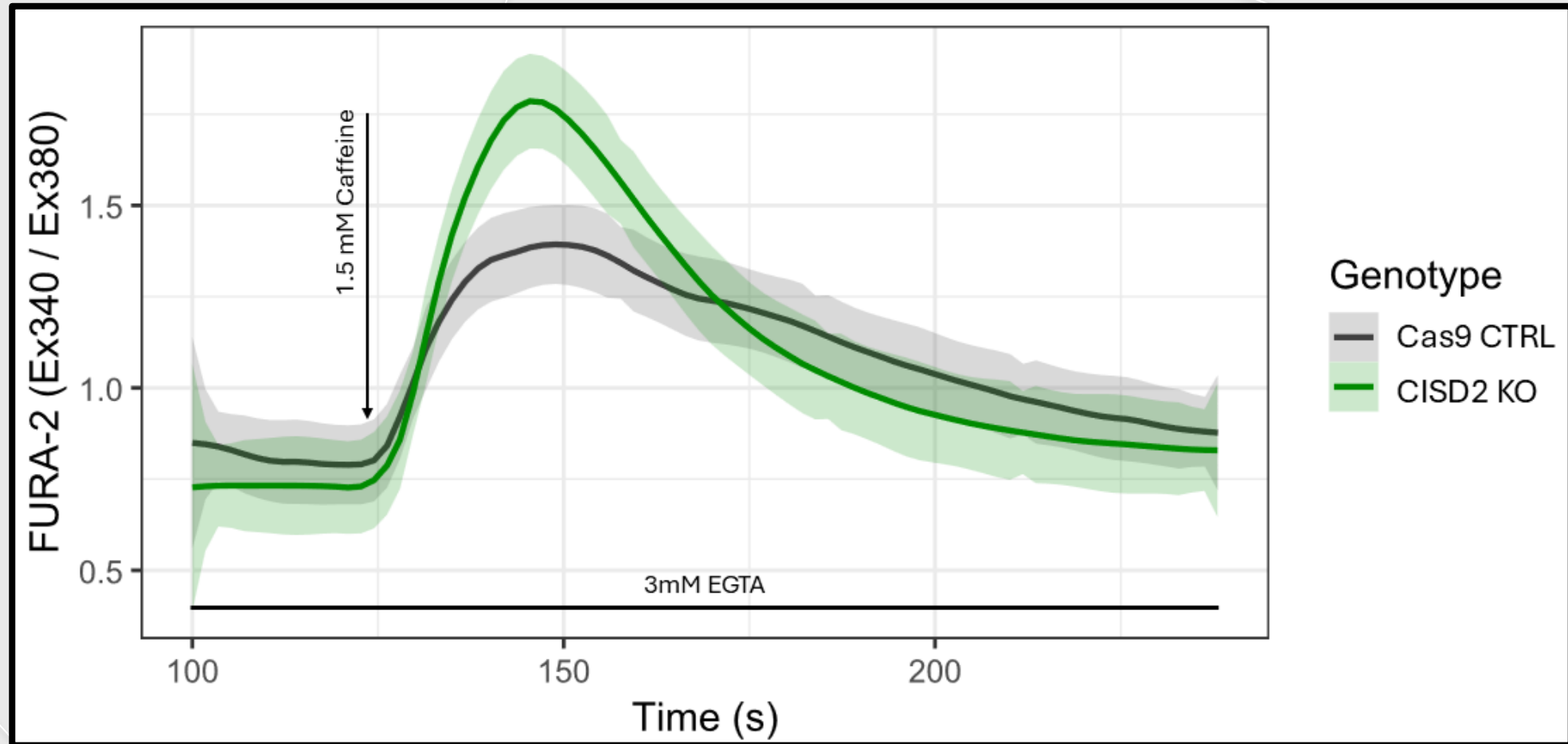
# Results

# Co-precipitation of RyR3 and CISO2

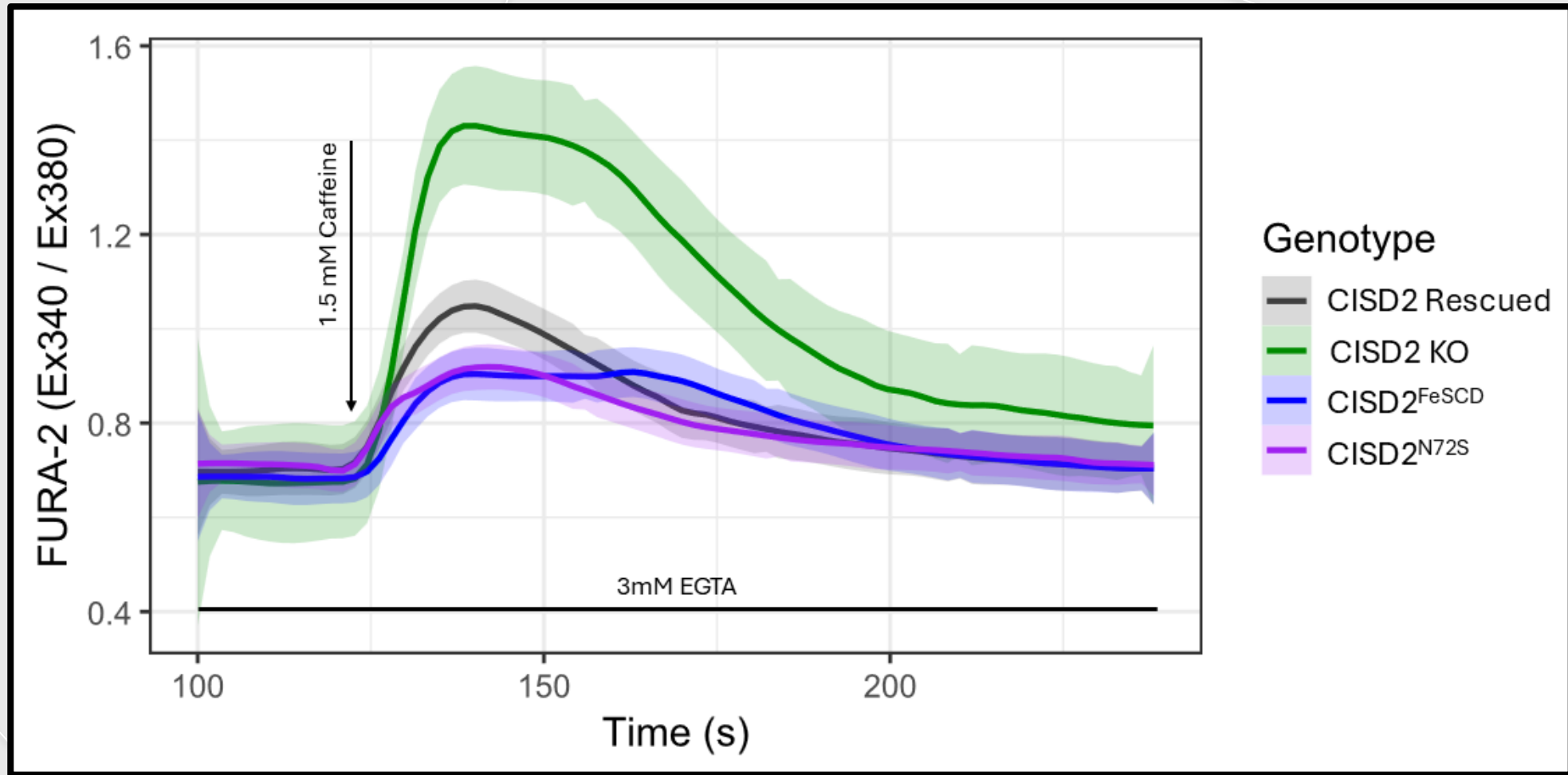
- RyR3 bands detected with C34 anti RyR3 antibodies
- CISO2 bands detected with anti CISO2 antibodies
- CISO2 band represents the amount bound to RyR3



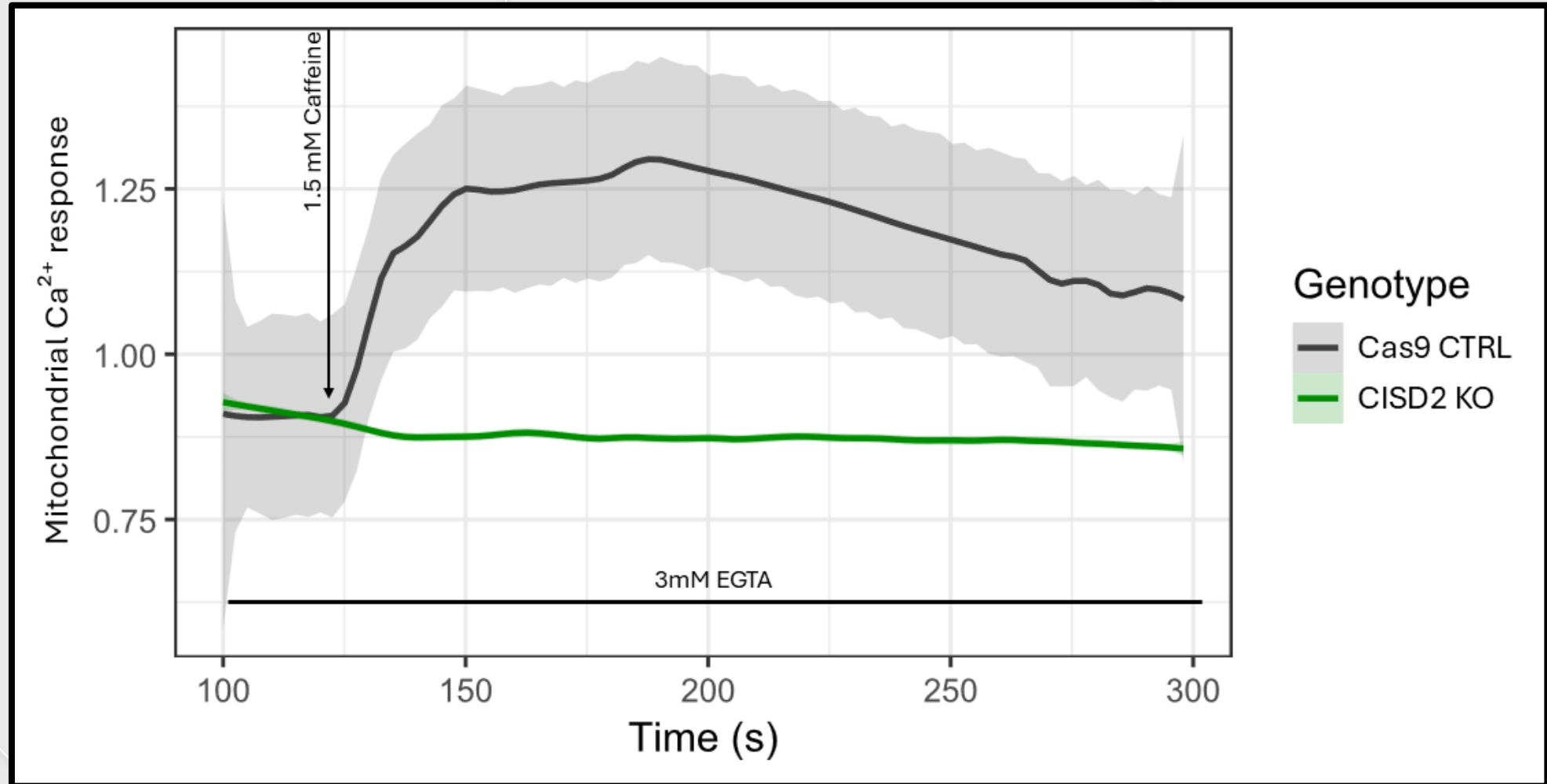
# Impact of C1SD2 on RyR3 activation



# Impact of CISO2 mutants on RyR3 activation



# Evaluation of mitochondrial $\text{Ca}^{2+}$ level



# Conclusions

- ✓ CISO2 is physically linked to RyR3
- ✓ CISO2 has act as a negative regulator of RyR3  $\text{Ca}^{2+}$  release
- ✓ CISO2<sup>FeSCD</sup> and CISO2<sup>N72S</sup> mutants do not have the same effect as the absence of CISO2
  - ✓ In absence of CISO2  $\text{Ca}^{2+}$  does not enter the mitochondria

# Future aspects

- Reverse Co-Immunoprecipitation
- Look for C1SD2 mutants impact on other components or pathways
- Increase the number of replicates of mitochondrial  $\text{Ca}^{2+}$  studies

**Thank you for the attention**